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	Understanding and characterizing phase change processes is a daunting task in the thermal management of satellites, power systems for extended missions, and fluid control in the engines or tanks of space platforms. Boiling is a phenomenon that involves a complex phase change, resulting in the formation of dynamic bubbles. Dynamic bubbles exhibit distinct behavior under microgravity, where gravitational force significantly influences fluid dynamics, resulting in unpredictable performances in bubble dynamics. Past experiments by researchers have observed that gravity strongly impacts the dry-out of the heated plate, significantly reducing heat transfer coefficients and critical heat flux in microgravity conditions. In standard gravity, buoyancy plays a crucial role in removing vapor bubbles from the vicinity of the heater surface. However, under microgravity, bubbles tend to grow and coalesce instead of detaching from the heated surface, leading		

	to a boiling crisis. Therefore, it is imperative to comprehend the phase distribution and monitor temperature and pressure changes within the reservoirs to prevent catastrophic events. Predicting boiling heat transfer remains a challenging task, due to various mechanisms involved, such as bubble nucleation, evaporation at the contact line, interactions, coalescences, and sliding.
Task Description:	Given various bubble mechanisms, we hypothesize that two key factors will predominantly impact bubbles: evaporation under microgravity and liquid agitation during bubble-bubble interactions. Testing this hypothesis requires a rigorous quantification of bubble features. Understanding the connection between boiling and bubble dynamics for different gravity conditions is crucial, but advancements have been challenging due to the intrinsically complex, high-dimensional, and dynamic nature of bubbles. With recent progress in artificial intelligence, computer vision, and scientific machine learning models (ML), they can now be integrated into boiling research. Therefore, the overarching objective of this project is to gain fundamental insights into boiling physics by linking bubble statistics, heat transfer performance, and boiling theory across various gravity conditions. To achieve this goal, we are developing a data-driven framework that constructs predictive models by consolidating datasets, extracting features, and leveraging ML methods. The proposed framework aims to identify crucial features contributing to boiling theory, especially for heat partitioning models and boiling curves under microgravity conditions. The success of this framework will provide a blueprint for designing dynamic bubbles, even in the presence of microgravity or reduced gravity. The new knowledge gained will be critical for developing two-phase thermal control systems for both manned and unmanned NASA missions in the future. This impact will bring significant societal benefits to thermal management solutions for space applications.
Rationale for HRP Directed Research:	
Research Impact/Earth Benefits:	Beyond its relevance to space systems, this research provides direct benefits to life on Earth by advancing the fundamental understanding of boiling heat transfer, a critical process in many energy and thermal management technologies. The insights gained from bubble dynamics, heat flux partitioning, and microlayer behavior will inform the design of more efficient cooling systems for high heat flux applications, including data centers, power electronics, and advanced manufacturing systems, ultimately reducing energy consumption and improving system reliability. This project also enables the development of innovative technologies, particularly in the integration of physics-informed machine learning with vision-based diagnostics. Such capabilities are especially impactful for next-generation thermal management systems, smart energy infrastructure, and autonomous monitoring of industrial processes. Collectively, this work contributes to more sustainable, efficient, and intelligent thermal systems across a wide range of terrestrial applications.
Task Progress:	Science Question Two-phase boiling improves thermal management for spacecraft technology, but the incomplete understanding of two-phase phenomena restrains technological deployment. Recent advancements in deep learning open new possibilities for investigating boiling heat transfer. Thus, the reinvestigation of microgravity datasets aboard the International Space Station (ISS) with deep vision techniques accelerates the fundamental understanding and deployment of boiling heat transfer in microgravity. Boiling heat transfer is the pinnacle of heat transfer efficiency but its highly transient nature requires accurate prediction to avoid system failure. Compared to single-phase, two-phase boiling produces higher heat transfer efficiency at constant temperature conditions. Though two-phase boiling is highly efficient, it poses an innate risk due to its highly transient nature. Microgravity magnifies the uncertain nature of boiling, where the lack of convection fundamentally alters nucleation cycles. Under microgravity, bubbles do not cyclically nucleate and depart but remain attached to the heater for extended periods of time. As heat flux increases, vapor covers a large portion of the heater, impeding heat transfer. With increasing heat flux and decreasing heat transfer, component failure occurs, known as the Critical Heat Flux (CHF). Thus, the prediction of boiling evolution with applied heat flux is crucial to the deployment of efficient heat transfer systems in microgravity. To predict boiling, there are two main methods: empirical correlations and mechanistic models. Forming empirical correlations requires only experimental data. However, boiling correlations cannot generalize across system conditions such as pressure, geometry, and working fluid. Mechanistic models stem from the mechanisms of boiling, incorporating statistics that generalize across boiling conditions. No mechanistic physics-based model currently exists for microgravity application. In recent years, advancements in machine learning now enable a new perspective for collecting boiling statistics to form a mechanistic model. By revisiting legacy NASA datasets with recently enabled machine learning techniques, this project develops a boiling mechanistic model. To achieve this goal, the project executes four sequential tasks. Task one evaluates NASA dataset quality, diversity, and potential for neural network-based segmentation. Task two focuses on automated extraction of relevant transient boiling statistics. As boiling features are collected, task three analyses these statistics to form a mechanistic model. Lastly, task four validates our mechanistic model by proving its generalizability to reconstruct the boiling curves across system parameters. Task 1. Data Collection The data collection task evaluates the quality and suitability of several NASA legacy datasets for mechanistic model development. Evaluation depends on experimental quality and diversity. Specifically, poor data sync, imaging resolution, and unaccounted system parameters impede mechanistic model results. Filtering the data accordingly saves time and builds confidence in the analysis. Generally, this process reveals the necessity for future experimentation aboard the ISS with higher image and data resolution. Task 2. Autonomous Feature Extraction Autonomous feature extraction divides into two branches: feature identification and development of a neural network capable of feature extraction. Thus, this analysis focuses on a literature review of relevant bubble statistics and their associated implications on heat transfer. Following feature identification, a Convolution Neural Network (CNN) is fine-tuned to autonomously segment all relevant bubble statistics. The project may use several CNNs in conjunction to separate features. Task 3. New Heat Partitioning Models

	Once statistics of boiling features are collected, the project investigates how features evolve with heat flux and time. Noting key trends and revisiting fundamental heat flux partitioning, a microgravity mechanistic model is developed. Task 4. Reconstruction of Boiling Curves To confirm the validity of the mechanistic model, the model reconstructs diversified boiling curves from the sole input of bubble statistics. The project targets 20% Root Mean Squared Error (RMSE) as the benchmark for successful deployment. Effectively, this shows the mechanistic model created captures the boiling physics properly and confidently establishes the dominating heat transfer modes in microgravity boiling. Not only does this work establish a readily deployable model, it illuminates future work to enhance the dominant modes of microgravity boiling.
Bibliography Type:	Description: (Last Updated: 04/28/2026)
Abstracts for Journals and Proceedings	Kayal R, Suh Y, Chang S, Raj R, Won Y. "Bubble oscillations in microgravity: Patterns and dynamics. " 12th International Conference on Boiling and Condensation Heat Transfer (ICBCHT12), Cambridge, Massachusetts, June 14-17, 2026. Abstracts. 12th International Conference on Boiling and Condensation Heat Transfer (ICBCHT12), Cambridge, Massachusetts, June 14-17, 2026. , Jun-2026